

HAFEEZULLAH EHSAN

Machine Learning Engineer

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Professional Summary

Computer science graduate with hands-on experience developing and evaluating machine learning solutions, including an NLP text classifier and retrieval-augmented generation chatbot. Skilled in Python, scikit-learn, data preprocessing, model evaluation, and building end-to-end ML applications. Communicates technical findings clearly through data visualization and practical prototypes.

Technical Skills

- **Languages:** Python, SQL
- **ML/NLP:** Scikit-Learn, Naive Bayes, TF-IDF Vectorization, Sentence Transformers, FAISS (Vector Search), Retrieval-Augmented Generation (RAG), Model Evaluation (Precision, Recall, Confusion Matrix), Supervised Learning, Statistics, NumPy, Pandas
- **Frameworks & Tools:** Flask, Streamlit, REST APIs, LLM API Integration (Groq), Git, VS Code, Jupyter Notebook
- **Core Competencies:** Data Preprocessing & Cleaning, Data Visualization, End-to-End ML Workflow, Cross-Functional Collaboration, Problem Solving

EXPERIENCE

Avancod (Pvt) Ltd

Jul 2024 - Aug 2024

Data Science and ML Intern

Islamabad, Pakistan

- Performed data preprocessing and cleaning on raw datasets using NumPy and pandas to prepare structured inputs for supervised learning model training.
- Implemented and evaluated machine-learning algorithms, applying scikit-learn and statistical model evaluation techniques to assess performance.
- Created data visualizations to communicate patterns and insights from processed datasets to the team.

Forage (Accenture)

Feb 2024 - Mar 2024

Data Analytics and Visualization Job Simulation

Remote, United States

- Simulated a Data Analyst role advising a social media client, cleaning, modeling, and analyzing 7 datasets to identify content trends.
- Prepared a presentation deck to clearly communicate technical findings, key insights, and recommendations to stakeholders.

PROJECTS

Student Mental Health Support Chatbot

- Python, Flask, FAISS, Sentence-Transformers, Groq LLM API
- Designed and built a retrieval-augmented generation (RAG) chatbot that preprocesses a student Q&A knowledge base, retrieves relevant context, and answers student mental health queries.
- Generated semantic embeddings using Sentence-Transformers (all-mpnet-base-v2) and indexed them with FAISS for fast similarity-based retrieval over student Q&A data.
- Integrated a large language model via the Groq API to generate context-aware, conversational responses grounded in retrieved data.
- Developed a Flask backend to serve the chatbot through a web interface, connecting the retrieval pipeline, LLM inference, and frontend in a complete end-to-end system.

SMS Spam Classifier

- Python, scikit-learn, Streamlit
- Built a supervised machine learning text classification system in Python and scikit-learn to detect spam messages, achieving strong accuracy on held-out test data.
- Implemented TF-IDF vectorization to transform unstructured text messages into numerical features for model training and evaluation.
- Trained and evaluated a Naive Bayes classification model using precision, recall, and confusion matrix analysis to validate performance.
- Deployed the trained model as an interactive web application using Streamlit, enabling real-time spam detection through a user-friendly interface.

EDUCATION

Pak-Austria Institute of Applied Sciences and Technology

2025

Bachelor of Science, Computer Science

Certifications

- Associate Data Engineer – SQL, *DataCamp (Nov 2025)*
- Data Analytics and Visualization Job Simulation, *Forage / Accenture North America (Mar 2024)*
- Supervised Learning with scikit-learn, *DataCamp (August 2026)*
- Natural Language Processing (NLP) in Python-, *DataCamp (August 2026)*